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BATTERY PACK

Abstract

A battery pack is provided with a tape unit and battery cells disposed on both sides of the tape unit. The battery cells include a first battery cell group disposed on a first surface of the tape unit, and a second battery cell group disposed on a second surface of the tape unit, wherein the battery cells in the first battery cell group are arranged alternately with the battery cells in the second battery cell group with respect to a direction of the tape unit. The tape unit is folded in a zigzag manner to form a battery cell stack.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority and the benefit of Korean Patent Application No. 10-2024-0030907, filed on Mar. 4, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

[0002] One or more embodiments relate to a battery pack.

2. Description of the Related Art

[0003] Generally, secondary batteries are chargeable and dischargeable unlike primary batteries that cannot be recharged. Secondary batteries are used as energy sources for mobile devices, electric vehicles, hybrid vehicles, electric bicycles, and uninterruptible power supplies. Depending on the type of external devices to which they are applied, the secondary batteries may be used in the form of a single battery or a module in which batteries are connected and bundled into one unit.

[0004] Small mobile devices, such as mobile phones, may operate for a certain period of time with an output from a single battery. However, when a long-term operation or high-power operation is required, such as in the case of electric vehicles or hybrid vehicles that consume a lot of power, a module type including multiple batteries is preferred due to issues of output and capacity. An output voltage or output current may be increased with the number of built-in batteries.

SUMMARY

[0005] One or more embodiments include a battery pack including a plurality of battery cells, which is reliable even if dropped because a swelling space between the plurality of battery cells.

[0006] However, the technical problems to be solved by the present disclosure are not limited to the above-mentioned problems, and other problems not mentioned may be clearly understood by those skilled in the art from the description herein.

[0007] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the present disclosure.

[0008] According to one or more embodiments, a battery pack includes a tape unit and battery cells disposed on surface of the tape unit, wherein the battery cells include a first battery cell group disposed on a first surface of the tape unit and a second battery cell group disposed on a second surface of the tape unit, wherein the battery cells of the first battery cell group are arranged alternately with the battery cells of the second battery cell group with respect to a direction of the tape unit, and wherein the tape unit is folded in a zigzag manner to form a battery cell stack.

[0009] The tape unit may include a first layer formed in a rectangular shape and comprising a plurality of adhesive parts are attached to one surface of the first layer at regular intervals, wherein the plurality of battery cells are disposed at positions corresponding to the plurality of adhesive parts.

[0010] The first layer may include a 1st-1 layer forming one surface of the first layer and a 1st-2 layer that is in contact with the 1st-1 layer and forms the other surface of the first layer, wherein the second layer may be attached to the 1st-2 layer.

[0011] The 1st-2 layer may include a first surface and a second surface, an adhesive may be formed only on the first surface, and the first surface may be adhered to the 1st-1 layer.

[0012] The plurality of adhesive parts may include first adhesive parts, second adhesive parts, and a third adhesive part, wherein each of the first adhesive parts may include a first part formed to correspond to an edge of a surface of one of the battery cells and a second part formed to correspond to a central portion of a surface of the one of the battery cells, wherein openings are

formed in the 1st-2 layer, and each of the second adhesive parts may be formed in a shape and at a position corresponding to one of the openings.

[0013] The 1st-1 layer may include a first hole unit, and the 1st-2 layer may include a second hole unit, wherein the first hole unit and the second hole unit may be formed in a shapes and at positions corresponding to each other.

[0014] The second layer may include a plurality of first adhesive parts each of which includes a first part formed to correspond to an edge of a surface of one of the battery cells and a second part formed to correspond to a central portion of the surface one of the battery cells, a plurality of second adhesive part each including a double-sided tape, and a third adhesive part including a compressible member, wherein the second layer may be formed so that the plurality of the first and second adhesive parts are arranged in a row on the first layer at regular intervals in the direction of the tape unit.

[0015] The third adhesive part may be disposed only at one end of the second layer.

[0016] The first adhesive unit and the second adhesive unit may be arranged alternately in the direction of the tape.

[0017] The first, second, and third adhesive parts all may include an adhesive material on both sides thereof.

[0018] The second layer may include the first adhesive parts arranged in a row at regular intervals in the direction of the tape unit.

[0019] The second layer may include a plurality of adhesive parts each including a double-sided tape; and a plurality of adhesive parts each including a compressible member, wherein the second layer includes the plurality of parts including the double-side tape and the plurality of adhesive parts each including the compressible member are arranged alternately in the direction of the tape unit.

[0020] According to one or more embodiments, a battery pack includes a tape unit and battery cells disposed on surfaces of the tape unit, wherein the battery cells may include a first battery cell group disposed on a first surface of the tape unit and a second battery cell group disposed on a second surface of the tape unit, wherein the battery cells of the first battery cell group may be arranged alternately with the battery cells of the second battery cell group with respect to a direction of the tape unit, wherein the tape unit may be folded in a zigzag manner to form a battery cell stack, wherein the tape unit may include a first layer formed in a rectangular shape, a second layer comprising a plurality of adhesive parts attached to one surface of the first layer at regular intervals, wherein the plurality of adhesive parts includes first adhesive parts, second adhesive parts, and a third adhesive part, wherein each of the first adhesive parts may include a first part formed to correspond to an edge of a surface of one of the battery cells, a second part formed to correspond to a central portion of the surface of one of the battery cells, and the first part may be formed to have a gap on one side.

[0021] According to one or more embodiments, a method of manufacturing a battery pack, the method including arranging first battery cell groups at regular intervals in a jig, fitting a first surface of the tape unit on the jig and adhering the first surface to the first battery cell group, and disposing a second battery cell group on a second surface of the tape unit, wherein the second battery cell group is arranged alternately with the first battery cell group.

[0022] The method may further include, before arranging the first battery cell group in the jig: stacking a 1st-2 layer on a 1st-1 layer forming a first surface of the tape unit to form a first layer, and stacking a plurality of adhesive parts on the 1st-2 layer to form a second layer of the tape unit.

[0023] The fitting of the first surface of the tape unit in the jig may include coupling first and second hole units in the first layer of the tape unit to a protrusion in the jig.

[0024] Other aspects, features and advantages other than those described above will become apparent from the detailed description, claims and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0026] FIG. 1 is an exploded view of a battery cell according to embodiments;

[0027] FIG. 2 is a perspective view of the battery cell of FIG. 1 in an assembled state;

[0028] FIG. 3 is a diagram showing a process of manufacturing a tape unit and a plurality of battery cells adhered to the tape unit according to embodiments;

[0029] FIG. 4 is a diagram showing battery cells adhered to a tape unit and a folding structure of the tape unit according to embodiments;

[0030] FIG. 5 is a diagram showing a structure in which battery cells adhered to the tape unit according to embodiments are folded in a zigzag manner to form a battery cell stack;

[0031] FIG. 6 is a cross-sectional view taken along line A-A' of FIG. 5;

[0032] FIG. 7 is a diagram showing an exploded perspective view of a tape unit according to embodiments;

[0033] FIG. 8(A) is a top view of a first adhesive unit according to embodiments, and FIG. 8(B) is a side view of the first adhesive unit according to embodiments;

[0034] FIG. 9(A) is a top view of the second adhesive unit according to embodiments, and

[0035] FIG. 9(B) is a side view of the second adhesive unit according to embodiments;

[0036] FIG. 10(A) is a top view of a third adhesive unit according to embodiments, and

[0037] FIG. 10(B) is a side view of the third adhesive unit according to embodiments;

[0038] FIG. 11 shows an exploded perspective view of a tape unit according to other embodiments;

[0039] FIG. 12 shows an exploded perspective view of a tape unit according to other embodiments; and

[0040] FIG. 13 is a flowchart showing a method of manufacturing a battery pack according to embodiments.

DETAILED DESCRIPTION

[0041] Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” if preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list.

[0042] Hereinafter, a battery pack according to embodiments will be described in detail with reference to the accompanying drawings. The terms or words used in the specification and claims should not be construed as limited to their usual or dictionary meanings, but should be interpreted as meaning and concept consistent with the technical idea of the present disclosure based on the principle of definability that the inventor may appropriately define the concept of terms in order to explain his or her invention in the best way. Accordingly, the embodiments described in this specification and the configurations shown in the drawings are only some of the most preferred embodiments of the present disclosure and do not represent the entire technical idea of the present disclosure, thus, it should be understood that at the time of filing this application, there may be various equivalents and modifications that may replace them. Also, when used herein, “comprise, include,” and/or “comprising, including” refer to specifying the presence of mentioned features, numbers, steps, operations, members, elements, and/or groups thereof, and it does not exclude the

presence or addition of one or more other shapes, numbers, operations, members, elements and/or groups. Additionally, when describing embodiments of the present disclosure, “may” and “may be” may include “one or more embodiments of the present disclosure.”

[0043] Additionally, to facilitate understanding of the disclosure, the attached drawings may not be drawn to scale, but the dimensions of some components may be exaggerated. Additionally, like reference numbers may be assigned to like components in different embodiments.

[0044] The statement that two comparison objects are ‘the same’ means that they are ‘substantially identical.’ Therefore, substantially identical may include a deviation that is considered low in the art, for example, a deviation of less than 5%. Also, the term that a certain parameter is uniform in a certain region may mean uniformity from an average perspective.

[0045] Although first, second, etc. are used to describe various components, these components are of course not limited by these terms. These terms are only used to distinguish one component from another component, and unless specifically stated to the contrary, the first component may also be a second component.

[0046] Throughout the specification, unless otherwise stated, each component may be singular or plural.

[0047] The placement of any component on the “top (or bottom)” of a component or the “top (or bottom)” of a component means that any component is placed in contact with the top (or bottom) of the component and may mean that other components may be interposed between the component and any component disposed on (or under) the component.

[0048] Additionally, when a component is described as being “connected,” “coupled,” or “combined” to another component, it should be understood that the components may be directly connected or combined to each other, but another component may be “interposed” between the components or each component may be “connected,” “coupled,” or “combined” through other components. Additionally, when a part is said to be electrically coupled to another part, this includes not only a case when the parts are directly connected, but also a case when the parts are connected with other intervening elements.

[0049] When referring to “A and/or B” throughout the specification, this means A, B or A and B, unless specifically stated to the contrary. That is, “and/or” includes all or any combination of the plurality of listed items. When referring to “C to D” this means C or higher and D or lower, unless specifically stated to the contrary.

[0050] The terms used in this specification are for describing embodiments of the present disclosure and are not intended to limit the disclosure.

[0051] Hereinafter, a battery cell according to embodiments will be described with reference to FIGS. **1** to **10(B)**.

[0052] FIG. **1** shows an exploded view of a battery cell C according to embodiments, FIG. **2** is a perspective view of the battery cell C of FIG. **1** in an assembled state, FIG. **3** shows a manufacturing process of a tape unit and a plurality of battery cells adhered to the tape unit according to embodiments, FIG. **4** shows a folding structure of the battery cell and the tape unit adhered to the tape unit according to embodiments, FIG. **5** shows a structure in which battery cells adhered to the tape unit according to embodiments are folded in a zigzag manner to form a battery cell stack, FIG. **6** is a cross-sectional view taken along line A-A' of FIG. **5**, FIG. **7** shows an exploded perspective view of the tape unit according to embodiments, FIG. **8(A)** is a top view of a first adhesive unit according to embodiments and FIG. **8(B)** is a side view of the first adhesive unit according to embodiments, FIG. **9(A)** is a top view of the second adhesive unit according to embodiments and FIG. **9(B)** is a side view of the second adhesive unit according to embodiments, and FIG. **10(A)** is a top view of a third adhesive unit according to embodiments and FIG. **10(B)** is a side view of the third adhesive unit according to embodiments.

[0053] Referring to FIGS. **1** and **2**, the battery cell C includes an electrode assembly **10**, an accommodation portion A in which the electrode assembly **10** is accommodated, and a sealing

portion TS extending along an edge of the accommodation portion A to seal the accommodation portion A. The electrode assembly **10** may be formed in a roll shape by winding first and second electrode plates **11** and **12** arranged to face each other with a separator **13** therebetween or may be formed in a stack form by stacking a plurality of the first and second electrode plates **11** and **12** with a separator **13** therebetween. An electrode tab **15** forming a charge and discharge path may be connected to the electrode assembly **10**. The electrode tab **15** may include two electrode tabs **15** of different polarities electrically connected to the electrode assembly **10**, and the electrode tab **15** connected to the electrode assembly **10** within the accommodation portion A may be drawn out to outside of the accommodation portion through a front surface F of the accommodation portion A. [0054] The accommodation portion A in which the electrode assembly **10** is accommodated may be formed in a substantially rectangular parallelepiped shape. More specifically, the accommodation portion A may include the front surface F from which the electrode tab **15** protrudes, a rear surface R opposite to the front surface F, a pair of main surfaces M that connect the front surface F and the rear surface R and occupy a relatively large area, and a pair of side surfaces D that connect the front surface F and the rear surface R and occupy have a relatively small surface area. The main surfaces M may occupy the largest surface area among the surfaces of the accommodation portion A and may be formed with a larger area than the front surface F, the rear surface R, and the side surfaces D. The front surface F, the rear surface R, and the pair of side surfaces D may be connected to the main surfaces M at different corners along the main surface M of the accommodation portion A.

[0055] In embodiments of the present disclosure, the main surface M and the side surface D may be formed in pairs at positions facing each other. For example, the main surface M may include first and second main surfaces M1 and M2 facing each other. Among the surfaces connecting the front surface F and the rear surface R, a surface having a relatively large surface area may correspond to the main surface M or the side surface M with a large surface area, and among the surfaces connecting the front surface F and the rear surface R, a surface having a relatively small surface area may correspond to the side surface M or the side surface D with a small surface area. Throughout this specification, the side surface D may denote a side surface D with a small surface area and, in particular, unless mentioned as a large surface area side surface M, may refer to a small surface area side surface D, and the large surface area side surface M may refer to the main side surface M.

[0056] The battery cell C may further include the sealing portion TS formed along the edge of the accommodation portion A to seal the accommodation portion A. The accommodation portion A for accommodating the electrode assembly **10** and the sealing portion TS for sealing the accommodation portion A may be formed from an exterior material **20** continuously formed to surround the electrode assembly **10**. That is, the exterior material **20** may surround the electrode assembly **10** and form the accommodation portion A that accommodates the electrode assembly **10**, and the extra exterior material **20** remaining after forming the accommodation portion A may form the sealing portion TS that seals the accommodation portion A.

[0057] The exterior material **20** may include a flexible exterior material **20** such as a layered structure, and more specifically, a metal layer **20a**, such as an aluminum sheet and an insulating layer **20b**, such as a resin coating layer formed on both surfaces of the metal layer **20a**. At this time, the metal layer **20a** may be exposed to the outside through a cross-section where the exterior material **20** ends. As described below, the metal layer **20a** may be exposed through at edge of the sealing portion TS where first and second exterior materials **21** and **22** are coupled to face each other with the electrode assembly **10** therebetween.

[0058] In embodiments of the present disclosure, the exterior material **20** may include the first and second exterior materials **21** and **22** that are coupled to face each other with the electrode assembly **10** therebetween. The electrode assembly **10** may be positioned between the first and second exterior materials **21** and **22**, the first and second exterior materials **21** and **22** may be folded to

overlap each other through a folding portion **25** connecting the first and second exterior materials **21** and **22**. Parts that contact each other may be joined to each other along edge regions of the first and second exterior materials **21** and **22** by heat fusion or the like. Thus, an inner region of the first and second exterior materials **21** and **22** facing each other with the electrode assembly **10** therebetween may be formed as the accommodation portion A, and the edge region of the first and second exterior materials **21** and **22** joined to each other may be formed as a sealing portion TS. [0059] The sealing portion TS may be formed continuously along the edge regions of the first and second exterior materials **21** and **22** except for the folding portion **25**. More specifically, the sealing portion TS may include a terrace portion T extending in a front surface F direction of the accommodation portion A and a side sealing portion S extending in a side surface D direction of the accommodation portion A. The electrode tab **15** connected to the electrode assembly **10** within the accommodation portion A may be pulled out to outside of the accommodation portion A through the terrace portion T extending in the front surface F direction of the accommodation portion A. In addition, a protection circuit module may be seated on the terrace portion T, and the electrode tab **15** that is pulled out through the terrace portion T may be bent to be connected to the protection circuit module seated on the terrace portion T.

[0060] The side sealing portion S may include a main body Sb at a position corresponding to the accommodation portion A and a front-end portion Sa extending from the main body Sb to a position outside of the accommodation portion A. The side sealing portion S may be folded toward the accommodation portion A to reduce an area occupied by the entire battery cell C, and the side sealing portion S may be folded toward the accommodation unit A through a first folding part and a second folding part. That is, in the first folding part, the main body (Sb) of the side sealing portion S at a position corresponding to the accommodation portion A and the front-end portion Sa of the side sealing unit S at a position outside of the accommodation portion A are folded up together. After the first folding, a second folding may be performed where the front-end portion Sa is folded to be concavely recessed toward edges where the front surface F of the accommodation portion A and the terrace portion T come into contact with each other. Through the first and second folding, the main body portion Sb of the side sealing unit S is folded up toward the side surface D of the accommodation portion A and may be placed on the side surface D. In the side sealing portion S, the front-end portion Sa may form a dog ear shape that is concavely recessed toward the edge formed by the front surface F of the accommodation portion A and the terrace portion T.

[0061] At the edge of the side sealing unit S, the metal layer **20a** may be exposed through the cross-section where the exterior material **20** forming the side sealing portion S ends. Because the main body Sb of the side sealing portion S is folded onto the side surface D of the accommodation portion A, the metal layer **20a** exposed through the edge of the main body Sb may not protrude from the side surface D of the accommodation portion A. The front-end unit Sa of the side sealing unit S is folded to be concavely recessed toward the edge formed by the front surface F of the accommodation portion A and the terrace portion T. Thus, the metal layer **20a** exposed through the edge of the front-end portion Sa of the side sealing portion S may not protrude from the front surface F of the accommodation portion A. That is, the side sealing unit S may not protrude from the side surface D of the accommodation portion A and the front surface F of the accommodation portion A and may be folded in a shape that follows an outer shape of the accommodation portion A while the main body Sb of the side sealing portion S is folded onto the side surface D of the accommodation portion A and the front-end portion Sa of the side sealing portion S is folded toward the edge where the terrace portion T and the front surface F of the accommodation portion A contact each other.

[0062] In embodiments of the present disclosure, an outer shape of the battery cell C may generally follow the outer shape of the accommodation portion A, and the outer surface of the accommodation portion A, i.e., the front surface F, the rear surface R, the main surface M, and the side surface D of the accommodation portion A may substantially denote the outer surface of the

battery cell C, The battery cell C may further include the sealing portion TS that seals the accommodation portion with the electrode assembly **10** is accommodated therein. The sealing unit TS, that is, at least the side seal unit S, may be folded toward the side surface D and the front surface F of the accommodation portion A in order to minimize an area occupied by the battery cell C. Thus, the external shape of the battery cell C may generally follow that of the accommodation portion A, and accordingly, the front surface F, the rear surface R, the main surface M and the side surface D of the accommodation portion A may substantially denote the front, back, main surface, and side surface of the battery cell C. The terrace portion T may be formed at a front position of the battery cell C, and the terrace portion T is generally in the form of a plate rather than a surface shape so that the protection circuit module is seated thereon. If the external shape of the battery cell C is viewed as approximately a rectangular parallelepiped shape, the front of the battery cell C may denote the front surface F of the accommodation portion A.

[0063] Referring to FIGS. **3** to **7**, the battery pack according to embodiments may include a tape unit **100** and a plurality of battery cells C disposed on both surfaces of the tape unit **100**. The plurality of battery cells C may include a first battery cell group CG1 disposed on a first surface **100a** of the tape unit **100** and a second battery cell group CG2 disposed on a second surface **100b** of the tape unit **100**. The battery cells C of the first battery cell group CG1 may be arranged alternately with the battery cells C of the second battery cell group CG2 with the tape unit **100** as the center. The tape unit **100** on which the first and second battery cell groups CG1 and CG2 are arranged may be folded in a zigzag manner to form a battery cell stack **1000**.

[0064] The tape unit **100** may include a first layer **110** integrally formed in a rectangular shape and a second layer **120** in which a plurality of adhesive parts **121**, **122**, and **123** are attached to one surface of the first layer **110** at regular intervals, wherein a plurality of battery cells C may be disposed at positions corresponding to the plurality of first, second, and third adhesive parts **121**, **122**, and **123**, respectively.

[0065] Typically, a tape for joining the battery cells C is disposed between a plurality of stacked battery cells. Because a plurality of tapes need to be individually attached between the plurality of battery cells, the speed of production of the battery pack may be reduced. Specifically, it is required to manage multiple tape materials separately, a jig is required each time an individual tape is attached between battery cells, and after one surface of the unit tape is attached to one surface of one battery cell, before attaching other battery cells to the other surface of the unit tape, the battery cell to which the unit tape is attached needs to be stored individually. Thus, the speed of production of the battery pack may be reduced.

[0066] Accordingly, in embodiments of this disclosure, an integrated tape unit **100** may be provided by attaching a plurality of adhesive parts **121**, **122**, and **123** of different shapes and different materials to the first layer **110**, which is an integrally formed single, rectangular tape.

[0067] The plurality of adhesive parts **121**, **122**, and **123** of different shapes and different materials may be easily attached to a plurality of battery cells C at one time using a jig J. Compared to a case where the tape is attached to each individual product, the speed of production of the battery pack may be improved by attaching at one time a plurality of adhesive parts to a plurality of battery cells using first, second, and third adhesive parts **121**, **122**, and **123** attached to the integrated tape.

[0068] According to embodiments, the first layer **110** may include a 1st-1 layer **111** forming one surface of the first layer **110** and a 1st-2 layer **112** that is in contact with the 1st-1 layer **111** and forms the other surface of the first layer **110**. The second layer **120** is attached to the 1st-2 layer **112**.

[0069] The 1st-1 layer **111** may be a release paper. At this time, the 1st-1 layer **111** may include a polyethylene terephthalate (PET) material. For example, the first layer **110** may include an anti-bubble material to prevent bubbles from generating between the 1st-2 layer **112** and the 1st-1 layer **111**, and thus, compactness of the battery pack may be maintained.

[0070] The 1st-2 layer **112** may include a PET material, and the 1st-1 layer **111** may be attached

thereto. The 1st-2 layer **112** may form a main skeleton unit that is integrally formed in a rectangular shape so that the tape unit **100** has a longitudinal direction.

[0071] The 1st-2 layer **112** may include a first surface **1121** and a second surface **1122**. According to embodiments, the adhesive material is formed only to the first surface **1121** of the 1st-2 layer **112**, and the first surface **1121** may be adhered to the 1st-1 layer **111**. The first, second, and third adhesive parts **121**, **122**, and **123** (which will be described later) may be attached to the second surface **1122**.

[0072] In a manufacturing process, the 1st-1 layer **111** and the 1st-2 layer **112** are combined first to form the first layer **110**, and the first, second, and third adhesive parts **121**, **122**, and **123** may be adhered to the second surface **1122** of the 1st-2 layer **112** of the first layer **110**. As shown in the drawings, because an area of the second surface **1122** is greater than an area of the first, second, and third adhesive parts **121**, **122**, and **123**, the adhesive material is not applied to all parts of the second surface **1122**. Accordingly, the first, second, and third adhesive parts **121**, **122**, and **123** may be attached to the second surface **1122**, which does not include an adhesive component, by using the adhesive material formed on surfaces of the first, second, and third adhesive parts **121**, **122**, and **123**.

[0073] According to embodiments, the second layer **120** may include the first, second, and third adhesive parts **121**, **122**, and **123**. The first adhesive part **121** may include a first part **121a** formed to correspond to an edge of a surface of the battery cell C and a second part **121b** formed to correspond to a central portion of a surface of the battery cell C. The first part **121a** may be formed to be harder than the second part **121b**. That is, the first part **121a** may include a hard material, and the second part **121b** may include a soft material. For example, the first part **121a** may include hard rubber. The first part **121a** including hard rubber may absorb shock if the battery pack is dropped. The second part **121b** may include a compressible material. For example, the second part **121b** may include sponge. If swelling of one or more the battery cells C occurs, the swelling may be effectively absorbed by the second part **121b** including a sponge material. Thus, the first adhesive part **121** may include different materials for the first part **121a** and the second part **121b**. The first adhesive part **121** including the first part **121a** and the second part **121b** including different materials is disposed on the second layer **120** to form one single layer. With such a configuration, if the battery pack including the battery cell stack **1000** is dropped, the shock resulting from the fall is absorbed through the first part **121a** including a hard material, and if swelling of one or more of the battery cells C occurs, particularly the central portion of the surface of a battery cell C swells, the second part **121b** disposed at a position corresponding to the swelling portion absorbs the swelling of the battery cell C.

[0074] In addition, the first part **121a** for ensuring durability and the second part **121b** for absorbing swelling are formed only one layer between the battery cells C. Thus, an increase in volume of the battery pack may be prevented, and, accordingly, a compact battery pack structure may be manufactured.

[0075] A first opening **112a** may be formed in the 1st-2 layer **112**, and the second part **121b** may be formed in a corresponding shape at a position corresponding to the first opening **112a**. That is, the second part **121b** of the second layer **120** may directly contact the 1st-1 layer **111**, which is a release paper, through the first opening **112a** formed in the 1st-2 layer **112**. Therefore, the second part **121b**, which includes a compressible material like a sponge, may secure an additional compressible space through the first opening **112a**, and may absorb swelling to a greater extent when swelling of the battery cells C occurs.

[0076] According to embodiments, the first part **121a** may be formed to surround the second part **121b**. The first part **121a** may be disposed between the battery cells C at a position corresponding to the edge of the battery cells C. That is, the first part **121a** may be formed at the edge position of the battery cell C that may absorb the impact that the battery cells C receive when the battery pack is dropped on any part thereof. In addition, the second part **121b** is disposed in the center of the

surface of the battery cell C, where swelling easily occurs, and may effectively absorb swelling of the battery cell C.

[0077] According to the present embodiments, the first part **121a** and the second part **121b** may be arranged to be spaced apart from each other by a (second) gap G2. As the first part **121a** is pressed by pressure between the battery cells C, the first part **121a** may spread in a direction perpendicular to the pressing direction. Likewise, the second part **121b** may receive the swelling pressure of the battery cell C and spread in a direction perpendicular to the direction of the swelling pressure. If the first part **121a** and the second part **121b** spread in a direction perpendicular to the direction of received pressure, interference may occur between the first part **121a** and the second part **121b**. Thus, the shock absorption of the battery pack and the swelling absorption of the battery cells C may not be performed properly. Accordingly, the second gap G2 is formed between the first part **121a** and the second part **121b** to prevent interference between the two components when the first part **121a** and the second part **121b** spread in a direction perpendicular to the direction of pressure received from the battery cell C. The second gap G2 may be formed, for example, to be 1 mm or more. If the second gap G2 is less than 1 mm, the gap between the first part **121a** and the second part **121b** may be filled and mutual interference may occur.

[0078] According to embodiments, the first part **121a** may be formed to have a (first) gap G1 including a cutout portion **130** cut on one side. The second portion **121b** may spread in a direction perpendicular to the direction of swelling pressure from the battery cell C. As the second part **121b** spreads, the first part **121a** may receive pressure from the second part **121b** in a direction perpendicular to the direction of pressure from the battery cell C. If the first part **121a** is formed in a ring shape, and if pressure is applied from the center of the ring shape to the ring shape, the ring shape may not be able to withstand the pressure and the ring shape may be damaged. Accordingly, as the first gap G1 is formed by cutting one side of the first part **121a** (which has a ring shape), when the swelling pressure received by the second part **121b** from the battery cell C is transmitted to the first part **121a**, the first part **121a** may not be damaged. That is, the first gap G1 increases as pressure transmitted from the second part **121b** is absorbed by the first part **121a**, and only a slight deformation occurs in the first part **121a** to the extent that the first gap G1 is widened. In addition, as the second part **121b** is pressed from the swelling of the battery cell C, the second part **121b** is pushed into a space of the first gap G1, and the shape of the first adhesive part **121** may be functionally maintained.

[0079] According to embodiments, the 1st-1 layer **111** may include a first hole unit **111b** and the 1st-2 layer **112** may include a second hole unit **112b**, wherein the first hole unit **111b** and the second hole unit **112b** may be formed in corresponding positions and in shapes corresponding to each other.

[0080] A jig protrusion JP of the jig J may be formed in a shape corresponding to the first hole unit **111b** and the second hole unit **112b**. And the first hole unit **111b** of the 1st-1 layer **111** and the second hole unit **112b** of the 1st-2 layer **112** may be coupled by fitting into the jig protrusions JP of the jig J. In this way, the 1st-1 layer **111** and the 1st-2 layer **112** constituting the first layer **110** may be fixed to a set position of the jig J. In a state that the 1st-1 layer **111** and the 1st-2 layer **112** are fixed at the set position, the second layer **120** and the battery cell C may be attached to the set position of the first layer **110**. Thus, the battery cell C, the first layer **110**, and the second layer **120** may be precisely attached to each other at the set positions.

[0081] According to embodiments, the second layer **120** may include the first adhesive unit **121** including the first part **121a** formed to correspond to an edge of a surface of the battery cell C and a second part **121b** formed to correspond to a central portion of a surface of the battery cell C. The second layer **120** may also include a second adhesive unit **122** including a double-sided tape, and a third adhesive unit **123** including a compressible member. The second layer **120** may be formed in a structure in which the plurality of adhesive parts **121**, **122**, and **123** are arranged in a row on the first layer **110** at regular intervals. Both surfaces of the first, second, and third adhesive parts **121**,

122, and **123** may include an adhesive A.

[0082] Referring to FIG. 7, the third adhesive part **123** may be disposed at only one end of the ends of the second layer **120**. The surfaces on both sides of the third adhesive part **123** may include an adhesive material. The third adhesive part **123** may include a compressible member capable of absorbing swelling but not include a hard member (such as the first part **121a** of the first adhesive unit **121**). Therefore, if the tape unit **100** on which the first and second battery cell groups CG1 and CG2 are disposed is folded in a zigzag shape to form the battery cell stack **1000**, with the third adhesive part **123** may be positioned at only one of the ends of the second layer **120**, a battery cell C may be located only on one side of the third adhesive part **123**.

[0083] Referring to FIG. 7, the first adhesive part **121** and the second adhesive part **122** may be alternately arranged. In such a case, when the tape unit **100** is folded in the zigzag manner of FIG. 4 to form the battery cell stack **1000** of FIG. 5, the result is a structure in which one surface of one battery cell C is in contact with the first adhesive part **121** and the other surface of the battery cell C is in contact with the second adhesive portion **122**. Here, the second adhesive part **122** may be a double-sided tape structure that has a less thickness (see FIGS. 8(b), 9(b), and 10(b)) compared to the thicknesses of the structures of the first adhesive part **121** and the third adhesive part **123**.

[0084] In the described embodiment, shock is absorbed by the tape unit **100** when a battery pack is dropped and also swelling of a battery cell C is absorbed by the tape unit **100**. Further, the battery pack may be compact by having a reduced overall volume because of the tape structure, in particular because the second adhesive portion **122** has a small thickness.

[0085] FIG. 11 shows an exploded perspective view of a tape unit according to other embodiments.

[0086] Referring to FIG. 11, a second layer **120'** may include first adhesive parts **121'** arranged in a row at regular intervals. That is, in these embodiments, the second layer **120'** only includes the first adhesive parts **121'** and not a second adhesive part and a third adhesive part. In this way, when forming a battery cell stack through a zigzag folding, the first adhesive part **121'** is interposed on both sides of the battery cells C. Thus, there may be improved absorption of swelling from the battery cell C by the second part **121b'**, and absorption of an impact if the battery pack is dropped may be further improved because of the first portion **121a'**.

[0087] FIG. 12 shows an exploded perspective view of a tape unit according to other embodiments.

[0088] Referring to FIG. 12, a second layer **120''** may include second adhesive parts **122''** and third adhesive parts **123''** alternately arranged alternately. That is, in these embodiments, the second layer **120''** is configured with only the second adhesive part **122''** and the third adhesive part **123''** and not the first adhesive part. In this way, when stacking the battery cells C through a zigzag folding, the second adhesive parts **122''** and third adhesive parts **123''** are alternately disposed on both sides of the battery cells C, and a first part **121a** of the first adhesive part including a hard material (as described above) is not interposed between the battery cells C. Thus, the volume of the battery pack may be reduced such that the battery pack is more compact, and, at the same time, the third adhesive part **123''** may absorb swelling that occurs between battery cells C.

[0089] With reference to FIG. 13, a method of manufacturing a battery pack according to embodiments will be described. FIG. 13 is a flowchart showing a method of manufacturing a battery pack according to embodiments. The method of manufacturing a battery pack according to embodiments may include stacking a 1st-2 layer **112** on a 1st-1 layer **111** forming a first surface **100a** of a tape unit **100** (S100), stacking a plurality of the first, second, and third adhesive parts **121**, **122**, and **123** on the 1st-2 layer **112** and forming a second surface **100a** of the tape unit **100** (S200), arranging a first battery cell group CG1 at regular intervals in a jig J (S300), fixing a first surface **100a** of the tape unit **100** on the jig J and adhering it to the first battery cell group CG1 (S400), disposing a second battery cell group CG2 on the second surface **100b** of the tape unit **100** (S500), and folding the tape unit **100** such that battery cells of second battery cell group CG2 at alternatively arranged with battery cells of the first battery cell group CG1 (S600).

[0090] The fixing the first surface **100a** of the tape unit **100** on the jig J and adhering it to the first

battery cell group CG1 (S400) may include coupling first and second hole units **111b** and **112b** of the first layer **110** of the tape unit **100** to a jig protrusion JP of the jig J.

[0091] In this way, a plurality of adhesive parts may be attached to a plurality of battery cells C at one time through the plurality adhesive parts **121**, **122**, and **123** attached to the first layer **110**, which is an integrated tape. After a plurality of battery cells are attached to the integrated tape, a battery cell stack may be formed simply by folding the integrated tape in a zigzag manner, thereby improving the production speed of the battery pack.

[0092] The effects that may be achieved through the present disclosure are not limited to the above-described effects, and other technical effects not mentioned will be clearly understood by those skilled in the art from the descriptions herein.

[0093] While the present disclosure has been described above with limited examples and drawings, the present disclosure is not limited thereto. Various modifications and changes may be made by those skilled in the art to which the present disclosure pertains, within the scope of equivalency of the technical spirit of the present disclosure and the scope of the claims described below.

[0094] It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure.

Claims

1. A battery pack comprising: a tape unit; and a battery cells disposed on surfaces of the tape unit, wherein the battery cells include: a first battery cell group disposed on a first surface of the tape unit; and a second battery cell group disposed on a second surface of the tape unit, wherein the battery cells in the first battery cell group are arranged alternately with the battery cells in the second battery cell group with respect a direction of the tape unit, and wherein the tape unit is folded in a zigzag manner to form a battery cell stack.
2. The battery pack as recited in claim 1, wherein the tape unit includes: a first layer formed in a rectangular shape; and a second layer comprising a plurality of adhesive parts attached to one surface of the first layer at regular intervals, wherein the plurality of battery cells are disposed at positions corresponding to the plurality of adhesive parts.
3. The battery pack as recited in claim 2, wherein the first layer includes a 1st-1 layer forming one surface of the first layer, and a 1st-2 layer that is in contact with the 1st-1 layer and forms another surface of the first layer, and wherein the second layer is attached to the 1st-2 layer.
4. The battery pack as recited in claim 3, wherein the 1st-2 layer includes a first surface and a second surface, wherein an adhesive is only on the first surface, and wherein the first surface is adhered to the 1st-1 layer.
5. The battery pack as recited in claim 3, wherein the plurality of parts includes first adhesive parts, second adhesive parts, and a third adhesive part, wherein each of the first adhesive parts includes: a first part formed to correspond to an edge of a surface of one of the battery cells; and a second part formed to correspond to a central portion of the surface of the one of the battery cells, wherein openings are formed in the 1st-2 layer, and wherein the each of the second adhesive parts is formed in a shape and a position corresponding to one of the openings.
6. The battery pack as recited in claim 3, wherein the 1st-1 layer includes a first hole unit, and the 1st-2 layer includes a second hole unit, wherein the first hole unit and the second hole unit are formed in a shapes and at positions corresponding to each other.
7. The battery pack as recited in claim 2, wherein the second layer includes: a plurality of first adhesive parts each of which includes a first part that corresponds to an edge of a surface of one of

the battery cells and a second part that corresponds to a central portion of the surface of the one of the battery cells; a plurality of second adhesive parts each including a double-sided tape; and a third adhesive part including a compressible member, wherein the second layer is formed so that the first and second, adhesive parts are arranged in a row on the first layer at regular intervals in the direction of the tape unit.

8. The battery pack as recited in claim 7, wherein the third adhesive part is disposed only at one end of the second layer.

9. The battery pack as recited in claim 7, wherein the first adhesive unit and the second adhesive unit are arranged alternately in the direction of the tape unit.

10. The battery pack as recited in claim 7, wherein the first, second, and third adhesive parts each include an adhesive material on both sides thereof.

11. The battery pack as recited in claim 7, wherein the second layer includes the first adhesive parts arranged in a row at regular intervals in the direction of the tape unit.

12. The battery pack as recited in claim 2, wherein the second layer includes: a plurality of adhesive parts each including a double-sided tape; and a plurality of adhesive parts each including a compressible member, wherein the second layer includes the plurality of parts including the double-side tape and the plurality of adhesive parts each including the compressible member are arranged alternately in the direction of the tape unit.

13. A battery pack comprising: a tape unit; and a battery cells disposed on surfaces of the tape unit, wherein battery cells include: a first battery cell group disposed on a first surface of the tape unit; and a second battery cell group disposed on a second surface of the tape unit, wherein the battery cells in the first battery cell group are arranged alternately with the battery cells in the second battery cell group with respect to a direction of the tape unit, wherein the tape unit is folded in a zigzag manner to form a battery cell stack, wherein the tape unit includes: a first layer formed in a rectangular shape; and a second layer comprising a plurality of adhesive parts attached to one surface of the first layer at regular intervals, wherein plurality of adhesive parts includes first adhesive parts, second adhesive parts, and a third adhesive part, and wherein each of the first adhesive parts includes: a first part formed to correspond to an edge of a surface of one of the battery cells; a second part formed to correspond to a central portion of the surface of one of the battery cells, and the first part is formed to have a gap on one side.

14. A method of manufacturing a battery pack, the method comprising: arranging a first battery cell group at regular intervals in a jig; fitting a first surface of a tape unit on the jig and adhering the first surface to the first battery cell group; and disposing a second battery cell group on a second surface of the tape unit, wherein the second battery cell group is arranged alternately with the first battery cell group.

15. The method as recited in claim 14, further comprising: before arranging the first battery cell group in the jig: stacking a 1st-2 layer on a 1st-1 layer forming a first surface of the tape unit to form a first layer; and stacking a plurality of adhesive parts on the f 1st-2 layer to form a second layer of the tape unit.

16. The method as recited in claim 14, wherein the fitting of the first surface of the tape unit in the jig includes coupling first and second hole units in the first layer of the tape unit to a protrusion in the jig.
